

Assignment No 3

Aim: Write a Java program that demonstrates the overloading method to compute power and absolute of a number for various data types and utilizes the static method from Math class for the same operation.

Code:

```
import java.util.Scanner;

class PowerFunction {

    Scanner sc = new Scanner(System.in);
    int num1, num2;
    public void powerCal() {
        num1 = 2;
        num2 = 3;
        System.out.println("Power of 2^3 : " + Math.pow(num1, num2));
    }

    public void powerCal(int num1, int num2) {
        System.out.println("Enter the base (int): ");
        int n1 = sc.nextInt();
        System.out.println("Enter the exponent (int): ");
        int n2 = sc.nextInt();
        System.out.println("Power of " + n1 + "^" + n2 + " : " + Math.pow(n1, n2));
    }

    public void powerCal(double num1, double num2) {
        System.out.println("Enter the base (double): ");
        double n1 = sc.nextDouble();
        System.out.println("Enter the exponent (double): ");
        double n2 = sc.nextDouble();
        System.out.println("Power of " + n1 + "^" + n2 + " : " + Math.pow(n1, n2));
    }
}

class AbsoluteFunction extends PowerFunction {

    int num;
    public void absoluteCal() {
        num = -12;
        System.out.println("Absolute Number of " + num + " is : " + Math.abs(num));
    }
    public void absoluteCal(int num) {
        System.out.println("Enter the number (int): ");
        int n = sc.nextInt();
        System.out.println("Absolute Number of " + n + " is : " + Math.abs(n));
    }
    public void absoluteCal(double num) {
        System.out.println("Enter the number (double): ");
```

```

        double n = sc.nextDouble();
        System.out.println("Absolute Number of " + n + " is : " + Math.abs(n));
    }
}

public class MethodOverloading {
    public static void main(String[] args) {

        AbsoluteFunction obj = new AbsoluteFunction();

        obj.powerCal();
        obj.powerCal(2, 7);
        obj.powerCal(2.0, 2.3);

        obj.absoluteCal();
        obj.absoluteCal(-9);
        obj.absoluteCal(-9.8);
    }
}

```

Output:

```

C:\Program Files\Java\jdk-21\bin>java.exe
Power of 2^3 : 8.0
Enter the base (int): 12
Enter the exponent (int): 2
Power of 12^2 : 144.0
Enter the base (double): 2
Enter the exponent (double): 3.4
Power of 2.0^3.4 : 10.556063286183154
Absolute Number of -12 is : 12
Enter the number (int): -90
Absolute Number of -90 is : 90
Enter the number (double): -9.5
Absolute Number of -9.5 is : 9.5

Process finished with exit code 0

```